

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)			Ticks in boxes 3, 4 and 5 i.e $3 \times (1)$ Extra box ticked -1	3			3		
	(b)	(i)	I	8 min 20 s		1		1	1	
			II	6 min 40 s		1		1	1	
		(ii)		Arrival time of P waves: 00 : 16 : 00 (1) Time difference: 00 : 02 : 40 ecf (1) [unless P wave arrival time > 18:40:00] Distance to epicentre: $1.6 \pm 0.2 [\times 10^3 \text{ km}]$ ecf from Δt (1) [no ecf on $d > 10 \times 10^3 \text{ km}$] $1:40 \rightarrow d = 1$ $2:00 \rightarrow d = 1.2$ $d = 1.4, 1.6, 1.8$ in tolerance $3:20 \rightarrow d = 2$			3	3	3	
		(iii)		1 correct circle or arc $\rightarrow$ 1 mark 2 correct circles or arcs $\rightarrow$ 2 marks 2 correct circles or arcs + labelled X $\rightarrow$ 3 marks (even if BC is incorrect) 3 circles or arcs intersecting at (almost) a point $\rightarrow$ 3 marks (even without X)			3	3	2	
				Question 2 total	3	2	6	11	7	0